

Yaning (Jason) Xu

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EDUCATION

McMaster University

Hamilton, ON

M.Sc. Computer Science — Thesis Stream | GPA: 4.0 / 4.0

Sept 2024 – Present

- Thesis: Mitigating Filter Bubble Effects in Recommender Systems using LLM-based intervention agents.
- Comp & Software **Graduate Scholarship** — awarded for outstanding research performance.
- Relevant coursework: Human-Centered AI; Information-Theoretic Methods in Trustworthy ML; Foundations of Interactive Learning; Empirical Research Methods for SE.

University of Waterloo

Waterloo, ON

B.Math. Computational Mathematics, Combinatorics & Optimization, Joint Honours.

Sept 2018 – Aug 2022

- **President's Scholarship of Distinction** — granted for outstanding academic performance upon admission.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, R, C, HTML/CSS

ML / AI: PyTorch, HuggingFace Transformers, LightFM, scikit-learn

Frameworks & Tools: React, Vite, Node.js, Tableau, Figma, Git

Competencies: Recommender systems, LLM integration, full-stack prototyping, user study design, algorithm design, data analysis

RESEARCH & PROJECTS

Kaleid — LLM Intervention Agent for Filter Bubble Mitigation

Sept 2024 – Present

MSc Thesis Project | McMaster University

- Built a **full-stack news recommendation platform** using React/Vite and LightFM to serve personalized articles and track real-time user interactions.
- Integrated a GPT-4-based intervention agent that analyzes exposure diversity metrics and delivers targeted nudges to reduce filter bubble effects.
- Designed and implementing a **longitudinal user study** (~60–80 participants) to measure the causal impact of LLM feedback on content diversity and opinion awareness.

StoryBlocks — AI-Assisted Narrative Design for Data-Driven Storytelling

2025-2026

Research Project | Manuscript in Preparation

- Co-developed an **AI-assisted authoring system** enabling writers to structure non-linear narrative ideas into coherent, data-driven story flows.
- Engineered the React/Vite front-end and designed the narrative-block **rendering algorithm**; led manuscript writing.
- Conducted a 30-participant user study evaluating system usability and narrative coherence outcomes.

Recommender Systems & Filter Bubble Effects

2025

Research Project | McMaster University

- Developed a controlled news recommendation platform to study how algorithmic choices and user behavior jointly shape filter bubble formation with 12 participants.
- Implemented full **recommendation pipeline** (React/Vite front-end, SQL data layer, JavaScript interaction logging) and performed quantitative analysis of diversity and bias metrics.

WORK EXPERIENCE

McMaster University

Hamilton, ON

Graduate Teaching Assistant — COMPSCI 2C03 (Data Structures) & 3AC3 (Complexity) Sept 2024 – Present

- Lead **weekly tutorial sessions for 40+ undergraduates** covering data structures, graph algorithms, algorithm design strategies, and complexity analysis.
- Develop and present structured tutorial slides; mentor students through problem sets, strengthening both technical communication and class management skills.

Hengshui Hengchuang Engineering Materials

Hebei, China

Business Data Analyst

Jun 2023 – Feb 2024

- Applied **max-flow/min-cut graph optimization** in Python to redesign shipment routing, reducing material transportation bottlenecks across the supply chain network.
- Built Tableau dashboards and Python/Excel reports to surface customer patterns, industry trends, and pricing correlations, directly informing procurement strategy.